

A Machine Learning Model for the Prediction of Hessian Matrices based on Semiempirical Electronic Structure Theory

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Introduction

- The Hessian matrix is a target of interest, due to its high computational cost and its versatile applications.
- Atomistic Features, based on GFN2-xTB^[1] quantities, are used for the prediction of the target: Hessian matrix elements
- The features are permutation and rotational invariant to achieve a unique mapping onto the target.
- The zero point vibrational energy is used to quantify the precision of the model.

Overview of the ML Model

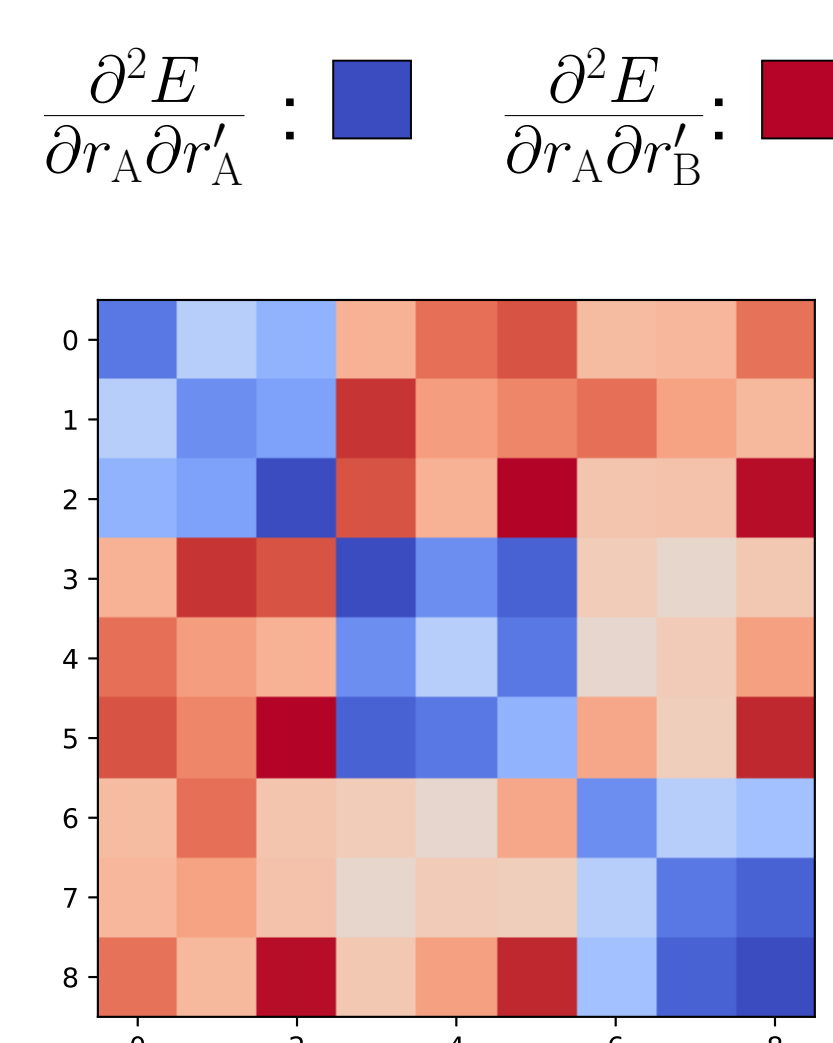
Origin of the Features

- Physics based, atomwise partitioned GFN2-xTB quantities are used for the features and were developed in our group and can be categorized into:

- Geometry: e.g. Coordination Number
- Density: e.g. Dipole Moment
- Energy: e.g. Coulomb-repulsion energy

The Hessian Matrix

- The elements of the Hessian are derivatives either in respect of the coordinates of a single atom or of two atoms
- For the ML model, we distinguish between the diagonal blocks (blue) and off-diagonal blocks (red)

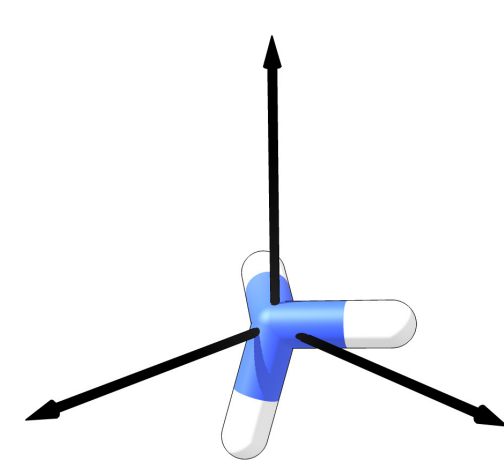
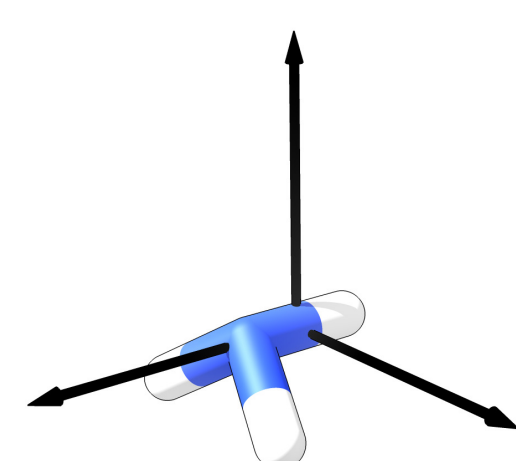


Connecting Features and Target

- The feature vector is either built up from the quantities of one or two atoms
- For ML modeling a constant feature vector length is necessary, hence two submodels are used which rely on ensemble based methods
- The features and targets are computed for each 3×3 block in a different orientation to achieve a consistent orientation

Model : Here the origin is shifted to the center of two atoms and the atoms are aligned to the z-axis.

Model : The origin of the coordinate system is shifted to the atom.

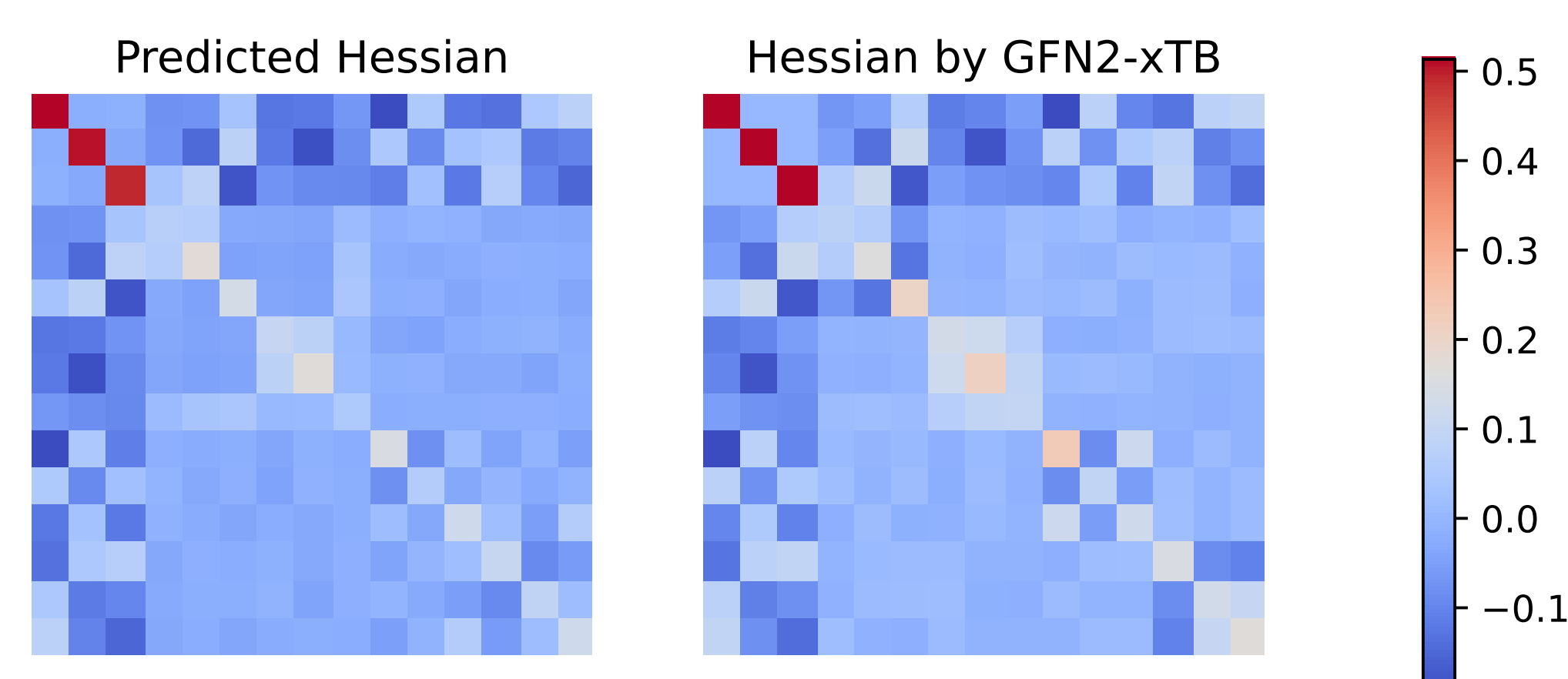
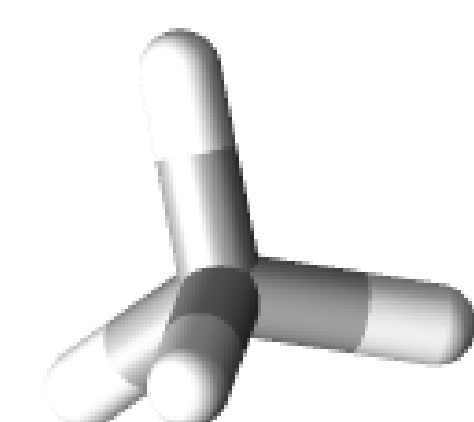


Unifying the Mapping of the Features

- An index is introduced to differ between the derivatives in respect to different coordinates of the same atom
- It is taken care of a consistent alignment to the atom oriented coordinate system

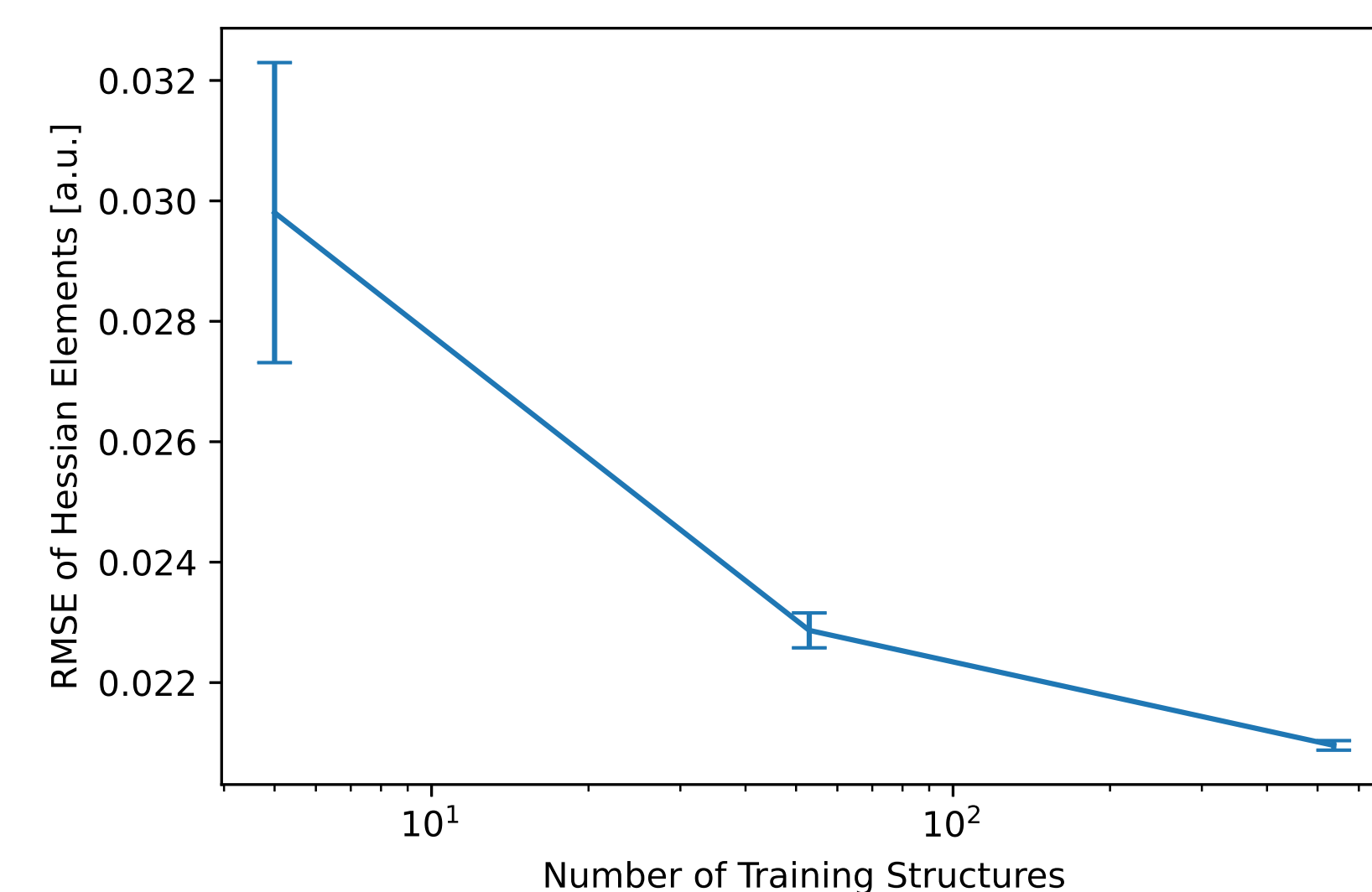
Comparison of predicted and GFN2-xTB Hessian

- Hessian matrix of methane as an example of the GDB7^[2] set
- Heatmaps are used to depict the Hessian matrices
- One can see the overall resemblance of the predicted Hessian compared to the computed one

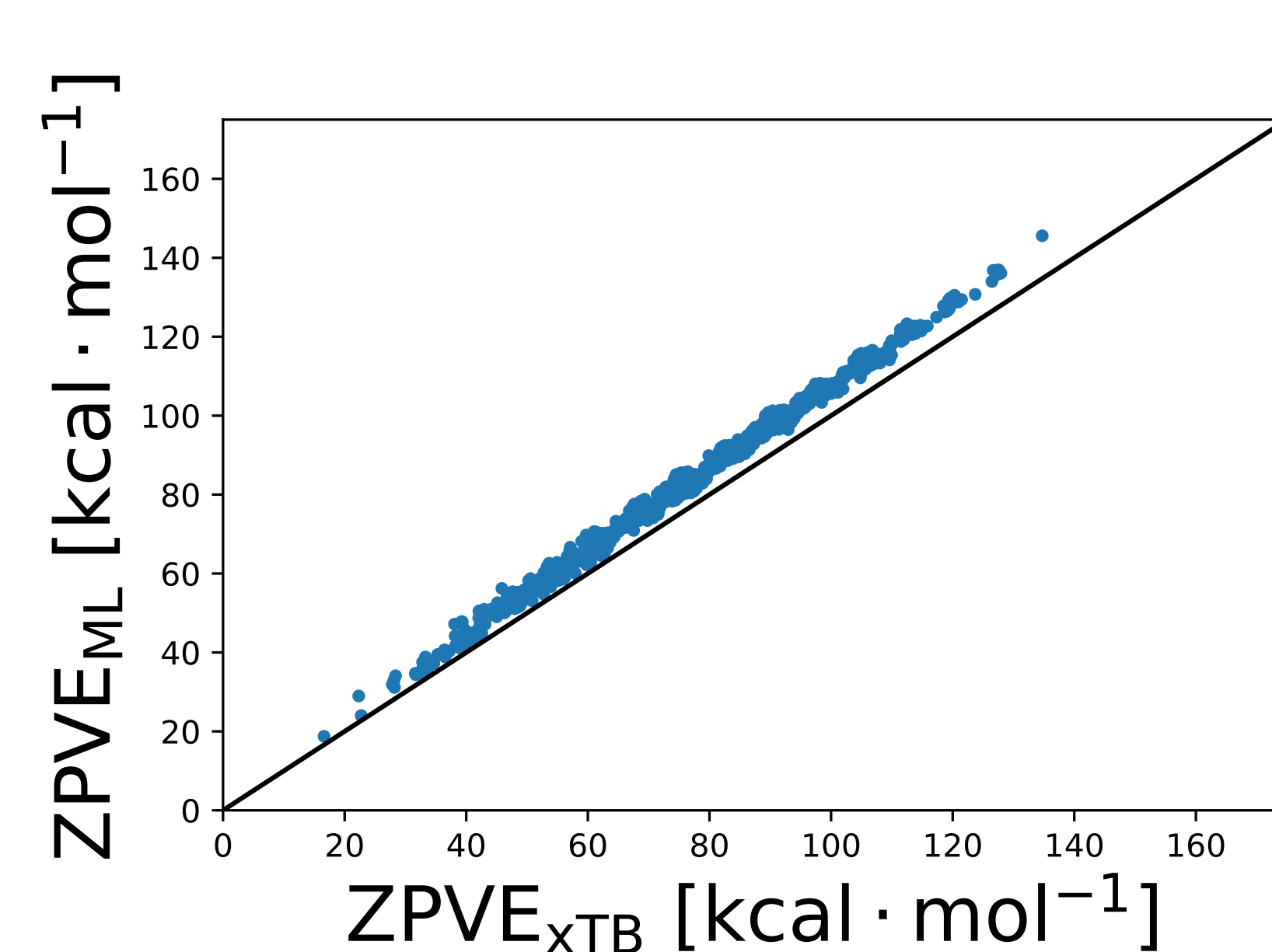


Predicted ZPVE of GBD7 Set

- Learning plot is computed using the GDB7^[2] set, which consists of 7164 structures
- The test set was fixed to 25 % of the size of the GBD7 set
- The training set was increased up to 15% of the



- Predicted zero point vibrational energy for the test set
- 15% of the full set were used for the training of the ML model
- One can see a low scattering of the data with a small systematic overestimation of the ZPVE



Conclusion and Future Perspective

- A ML model for the prediction of Hessian matrix elements with a verified learning behavior was developed
- This opens the opportunity to an accelerated computation of Hessian matrices, which can be embedded in computational workflows.
- Optimizing the feature target correlation for an enhanced prediction

References

- [1] C. Bannwarth, S. Ehlert, S. Grimme, *Journal of Chemical Theory and Computation* **2019**, 15, 1652.
- [2] M. Rupp, A. Tkatchenko, K.-R. Müller, O. A. von Lilienfeld, *Physical Review Letters* **2012**, 108.

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